FAG SmartCheck MAPS Template (Instrumentation)

The FAG SmartCheck MAPS template is designed to monitor some crucial values and alarms from a FAG SmartCheck for vibration monitoring on a motor and pump for example.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

FAG SmartCheck Functional Philosophy

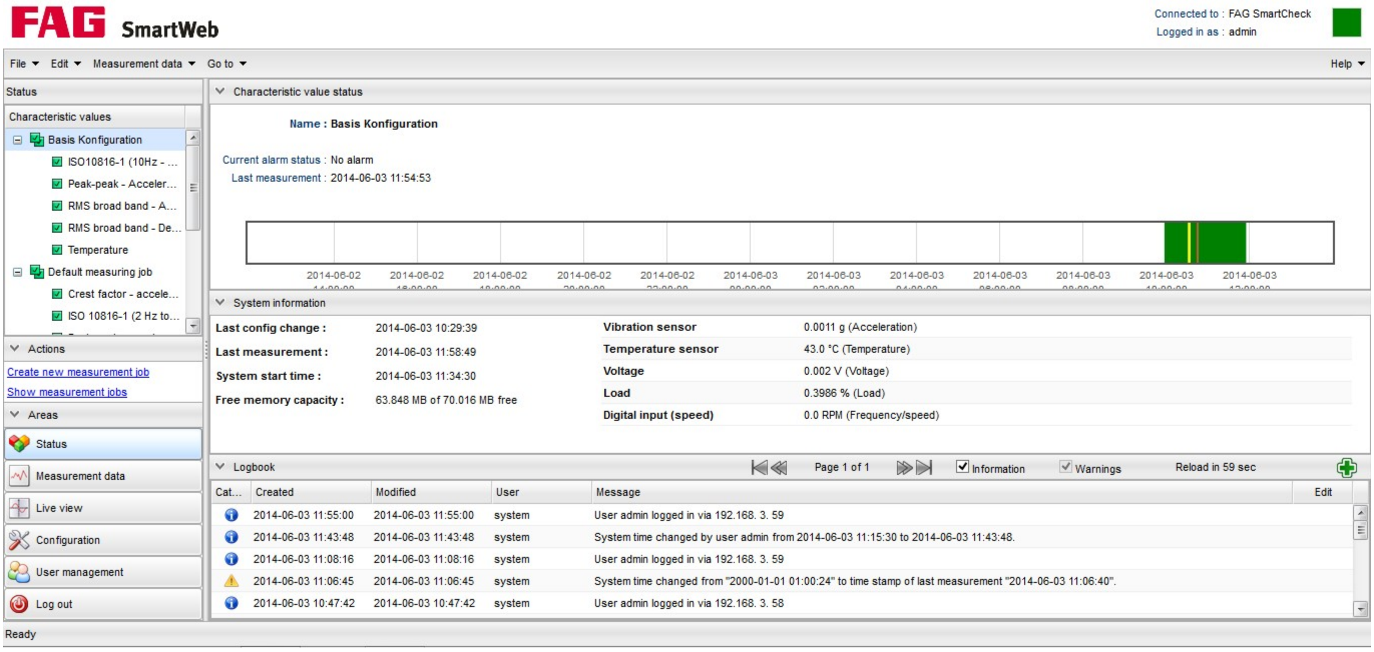
The FAG SmartCheck needs to be configured for communication to a Mitsubishi Q/L PLC (see below for instructions to achieve this setup). MAPS will create the PLC FB and SCADA faceplates automatically, but the user is required to enter the FAG SmartCheck start address into the current FB instance. This will also be show below. The FB will transfer the information as well as pack the Alarm values into separate addresses for SCADA use. The following signals can be used with this MAPS template and only these configured signals must be implemented on the FAG SmartCheck configuration.

config_version
communication_status
device_status
a_basis_konfiguration
a_iso10816_1_10hz_1khz_velocity
a_peak_peak_acceleration_high_v
a_rms_broad_band_acceleration_o
a_rms_broad_band_demodulation_o
a_temperature
v_iso10816_1_10hz_1khz_velocity
v_peak_peak_acceleration_high_v
v_rms_broad_band_acceleration_o
v_rms_broad_band_demodulation_o
v_temperature

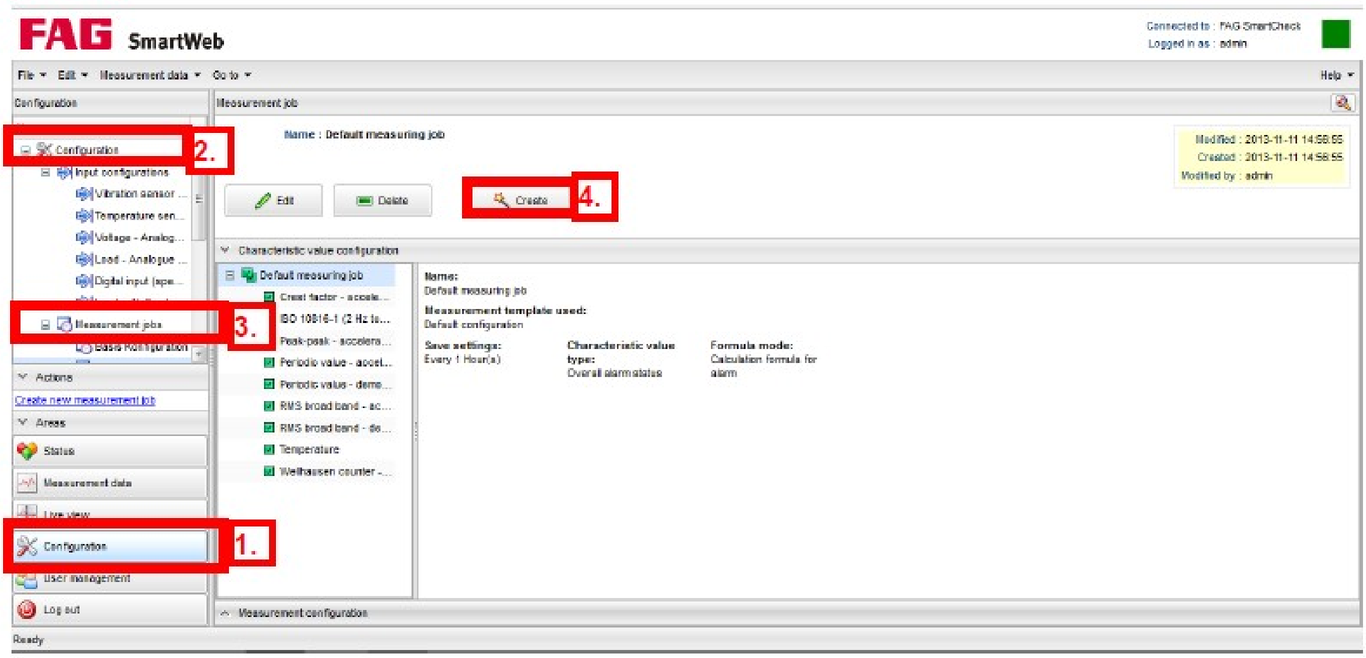
Setup of the FAG SmartCheck

Seting the Signals and measurements

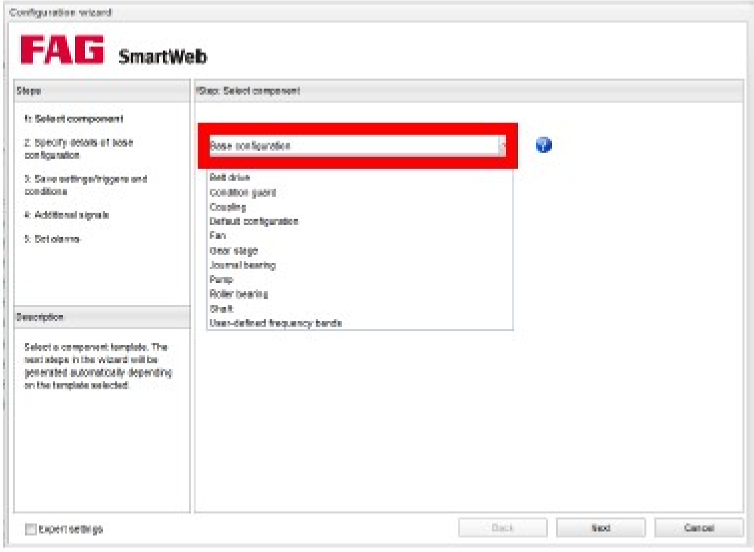
The configuration of the Smart Check is done by using a Standard Internet Browser. By entering the IP address of the Smart Check (default: 192.168.3.100)
After the loading of the web application the following screen appears:



Then select “Configuration” at “Areas” (“1.” in following screen), select “Measurement jobs” (“2.” and “3.”) and at last “Create” (“4”).

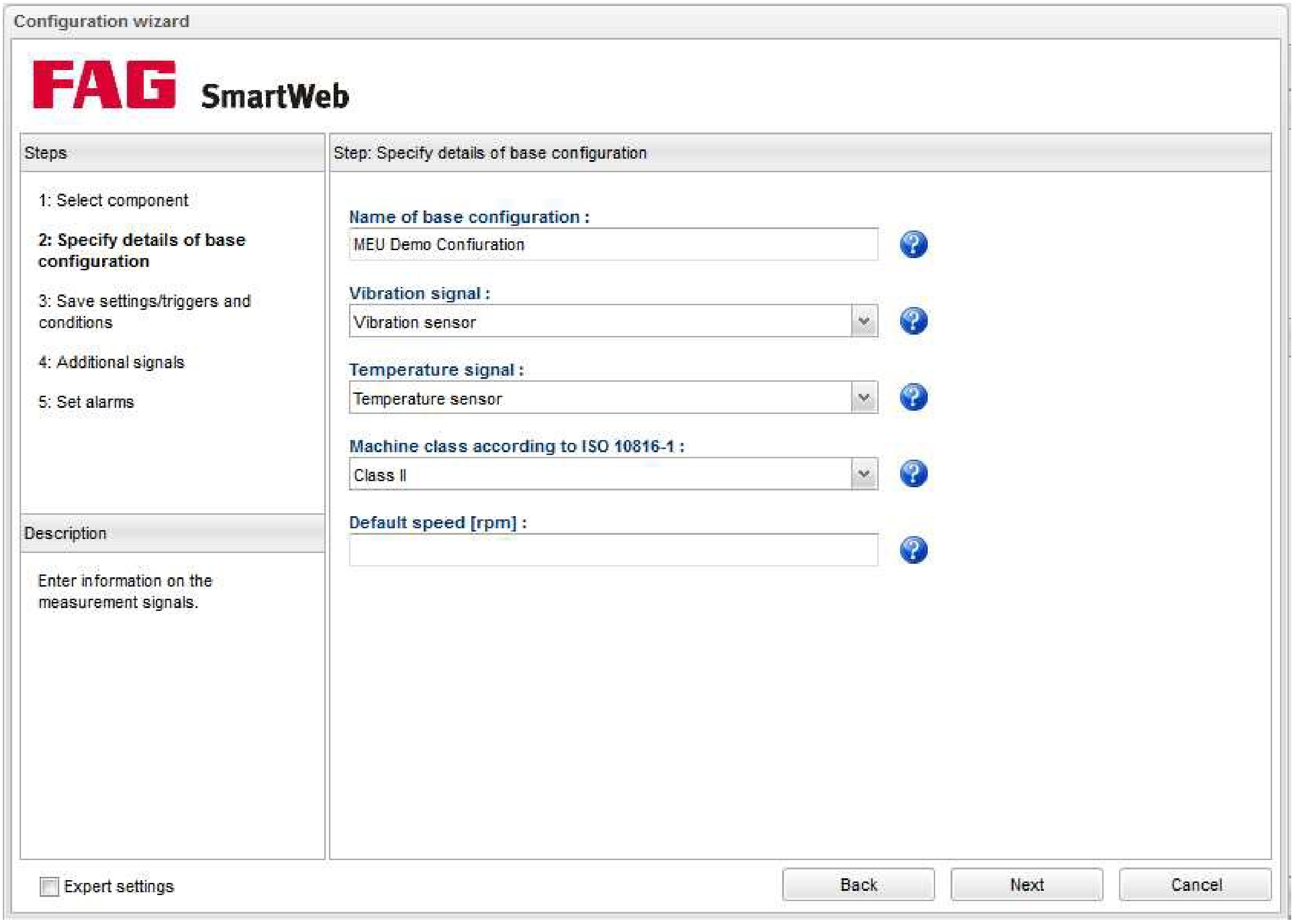


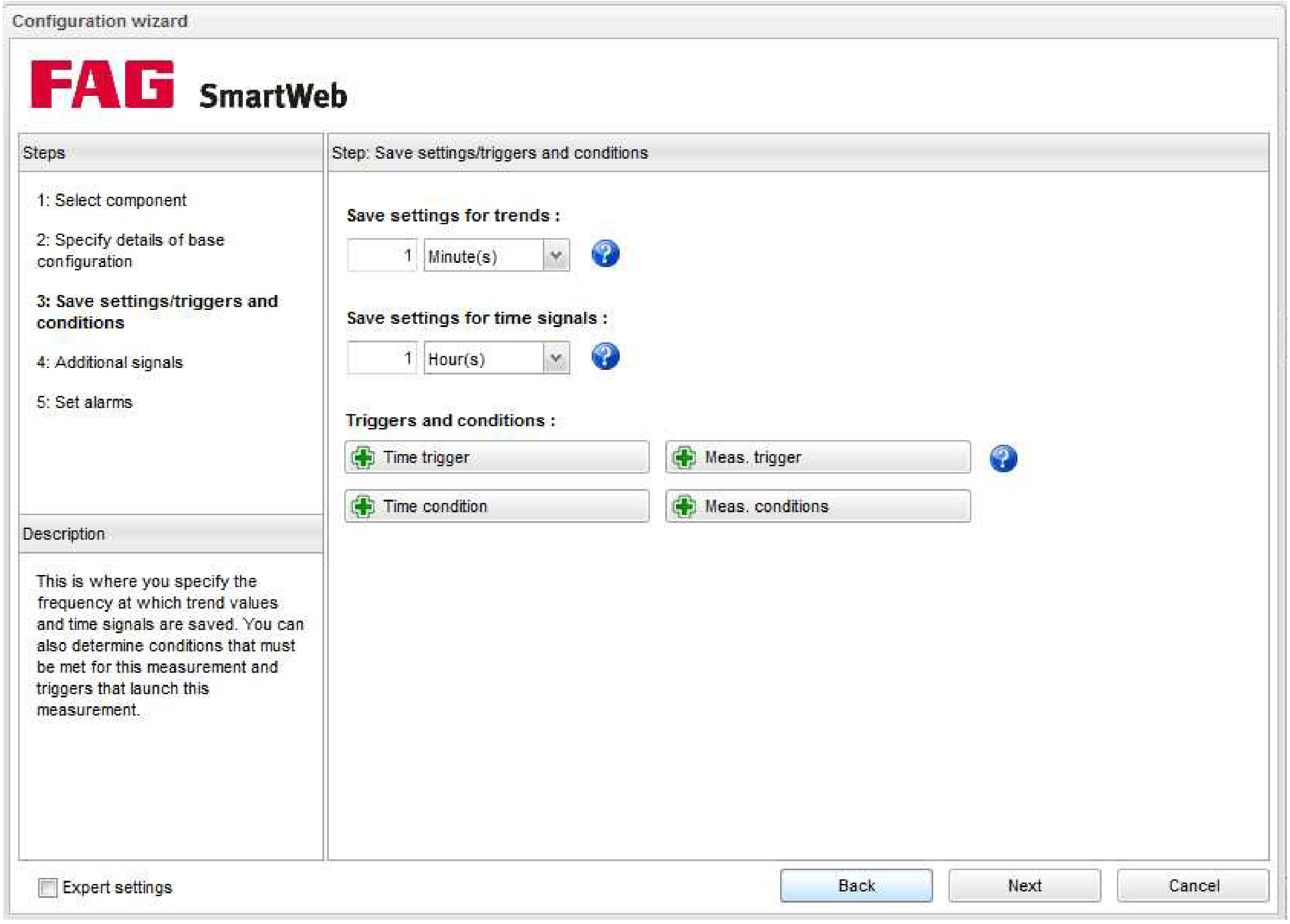
Select “Base Configuration” in the next screen.

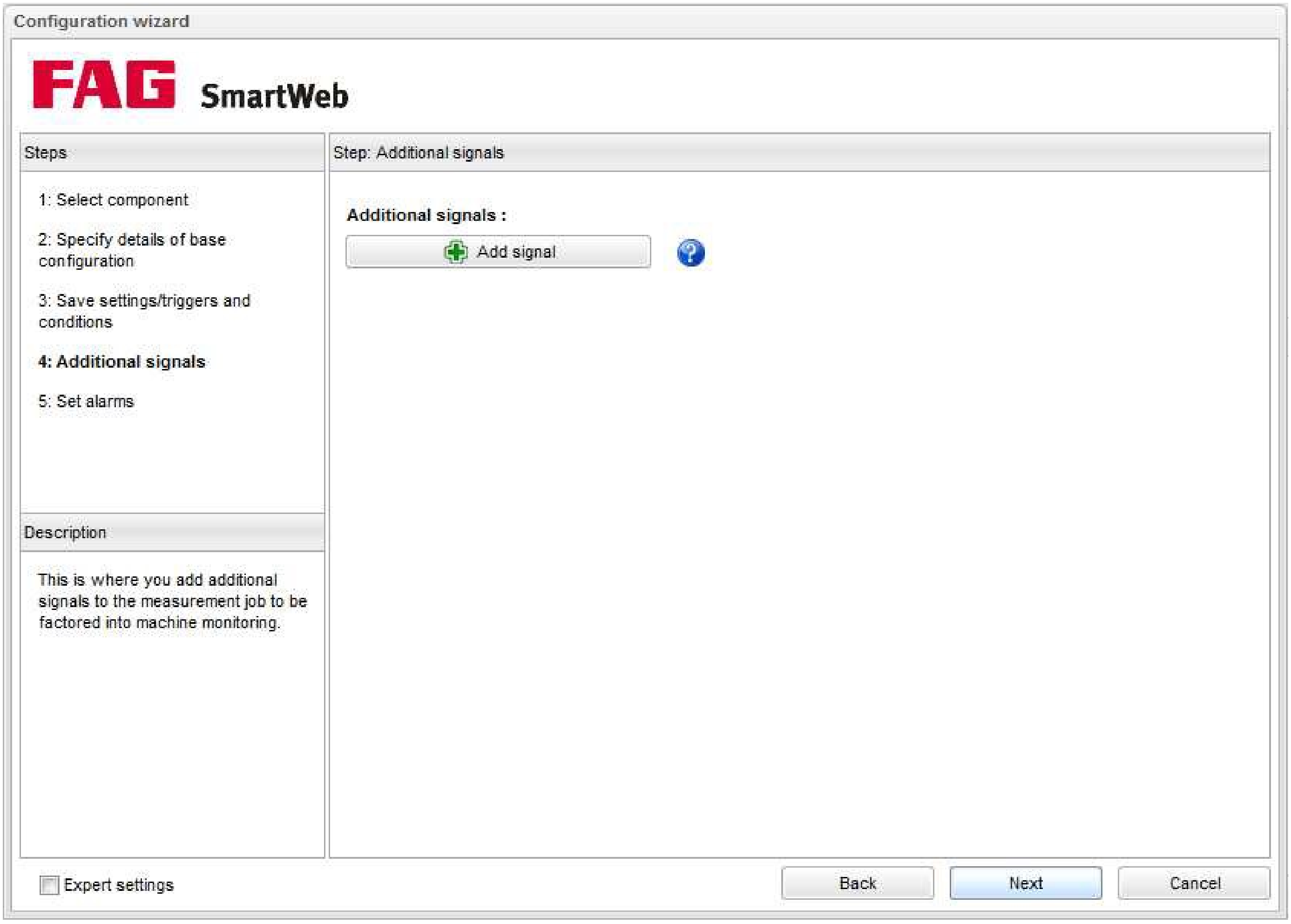


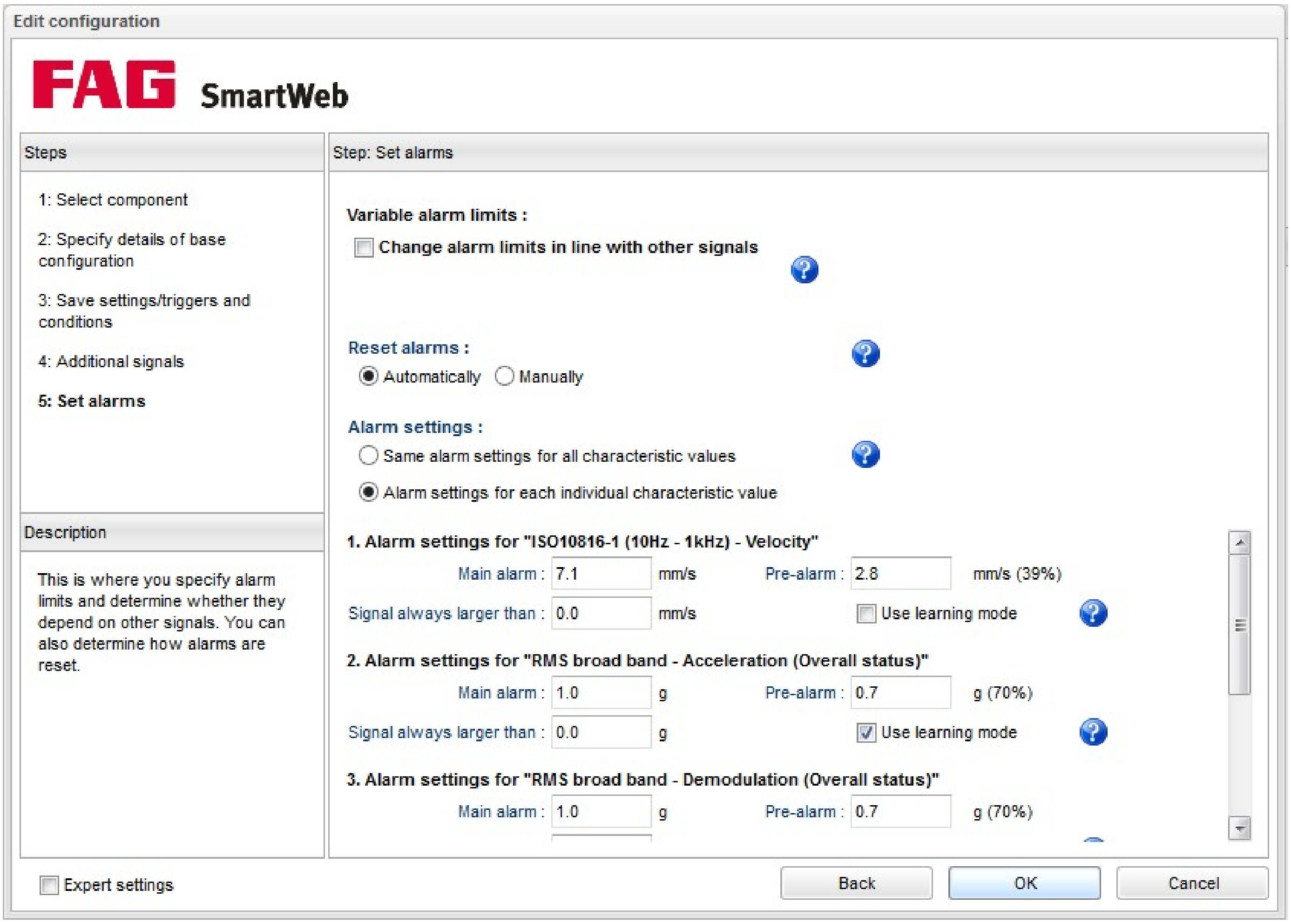
Following the wizard, you can define a new measurement job including the alarm settings.

Details about the settings can be found in the FAG SmartCheck manual.


Configuration wizard
FAG SmartWeb
Steps
1: Select component
2: Specify details of base configuration
3: Save settings/triggers and conditions
4: Additional signals
5: Set alarms
Description
Enter information on the measurement signals.
Step: Specify details of base configuration
Name of base configuration :
MEU Demo Confiuration
Vibration signal :
Vibration sensor
Temperature signal :
Temperature sensor
Machine class according to ISO 10816-1 :
Class II
Default speed [rpm] :
Expert settings
Back
Next
Cancel

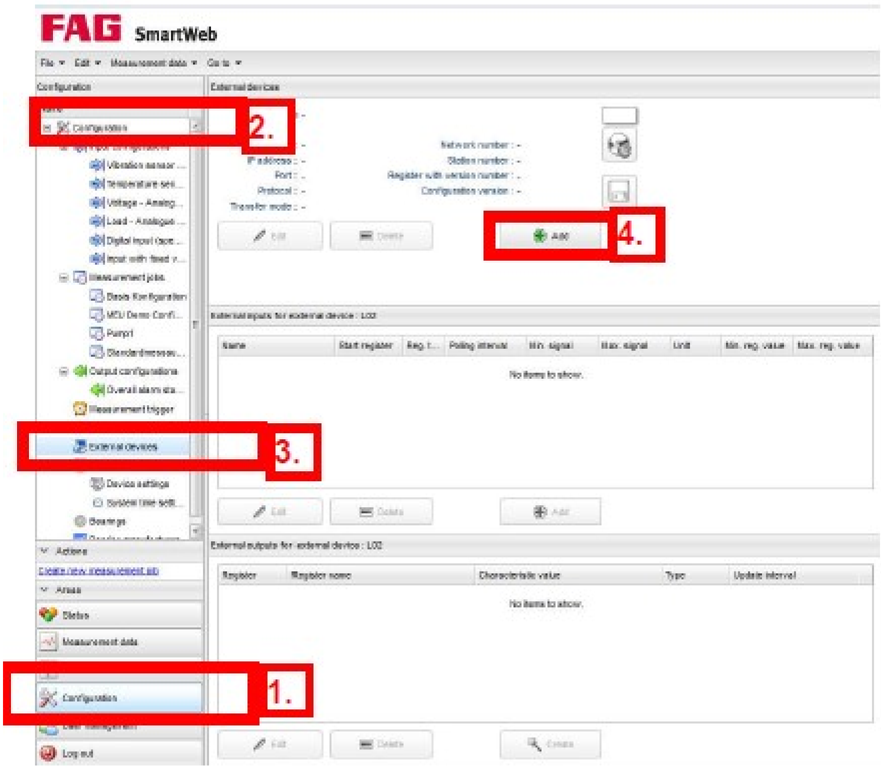

Configuration wizard
FAG SmartWeb
Steps
1: Select component
2: Specify details of base configuration
3: Save settings/triggers and conditions
4: Additional signals
5: Set alarms
Description
This is where you specify the frequency at which trend values and time signals are saved. You can also determine conditions that must be met for this measurement and triggers that launch this measurement.
Step: Save settings/triggers and conditions
Save settings for trends :
1
Minute(s)
Save settings for time signals :
1
Hour(s)
Triggers and conditions :
Time trigger
Meas. trigger
Time condition
Meas. conditions
Expert settings
Back
Next
Cancel


Configuration wizard
FAG SmartWeb
Steps
1: Select component
2: Specify details of base configuration
3: Save settings/triggers and conditions
4: Additional signals
5: Set alarms
Description
This is where you add additional signals to the measurement job to be factored into machine monitoring.
Step: Additional signals
Additional signals :
Add signal
Expert settings
Back
Next
Cancel



Setting the communications to the Mitsubishi Q/L PLC

Select "Configuration" at "Areas" ("1." in following screen), select "External Devices" ("2." and "3.") and at last "Add" ("4").



Specify the PLC settings as follows:

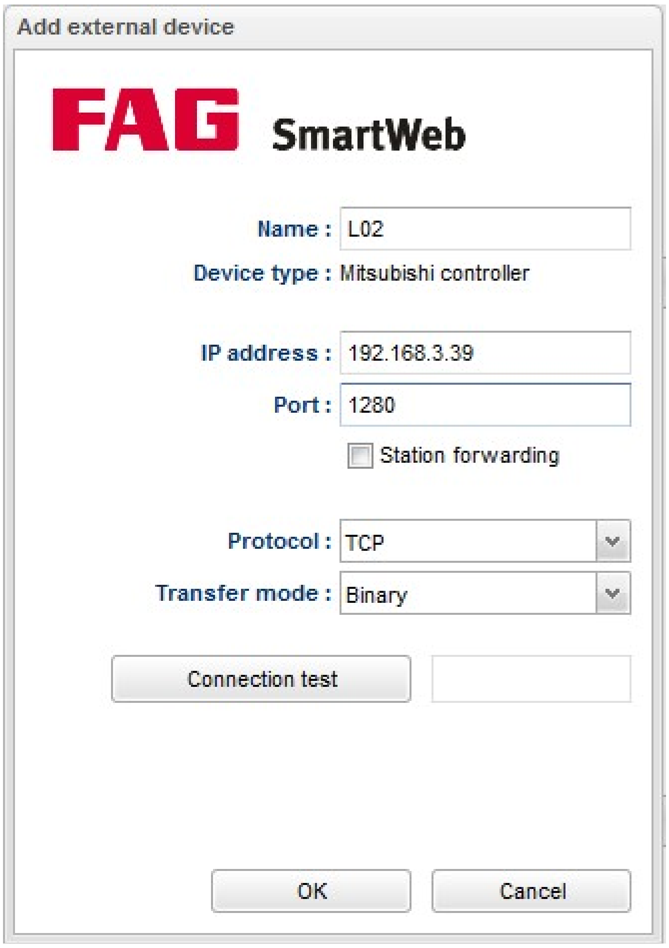
IP Address: CPU IP Address

Port: By default it is 1280, but if you use MAPS which is also set to 1280 by default it will not work, so in this case rather start using 1281 for SmartCheck 1 and 1282 for SmartCheck2 etc…

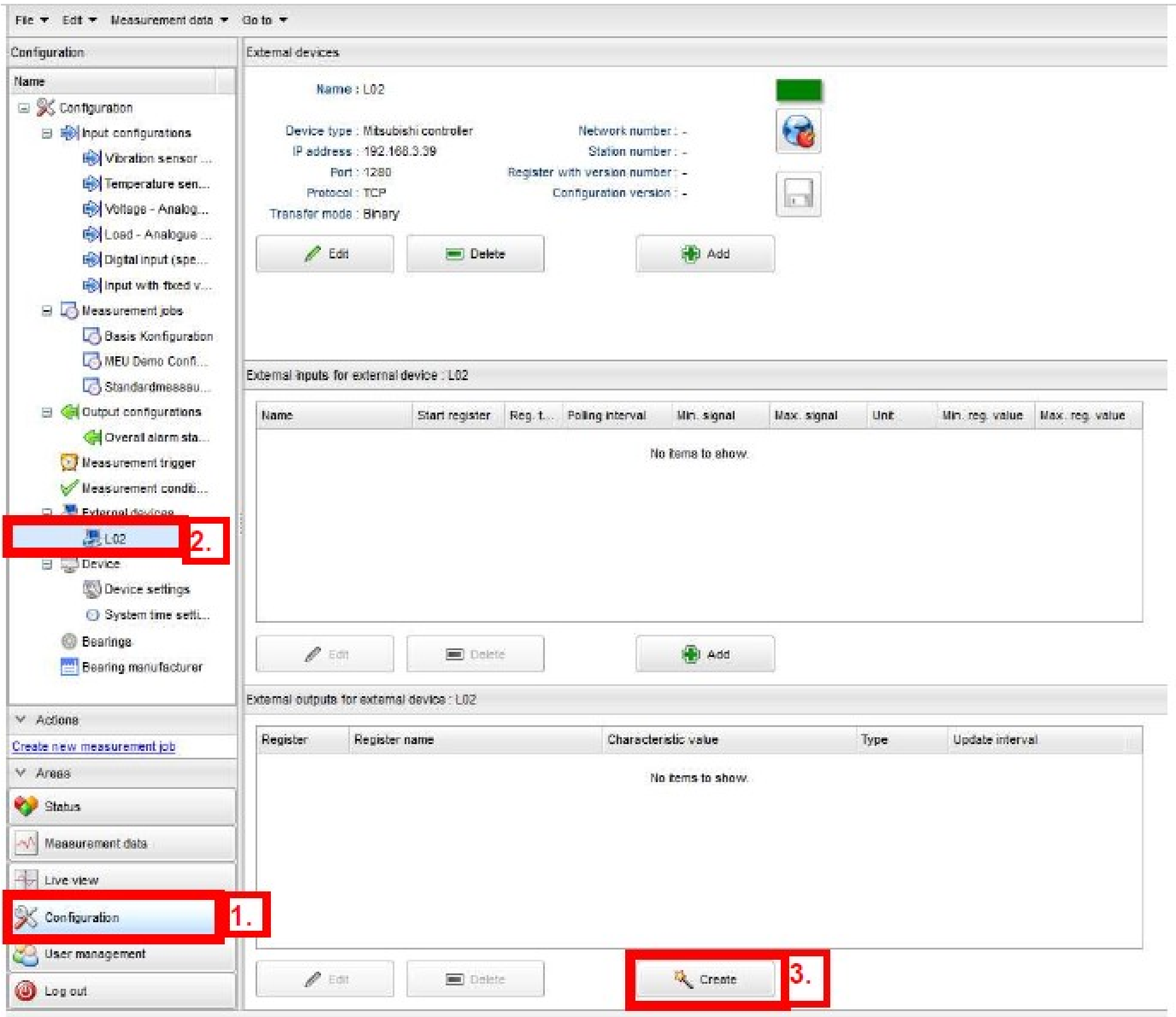
Protocol: TCP

Transfer Mode: Binary

With pressing “Connection Test“ button, the communication status can be checked.

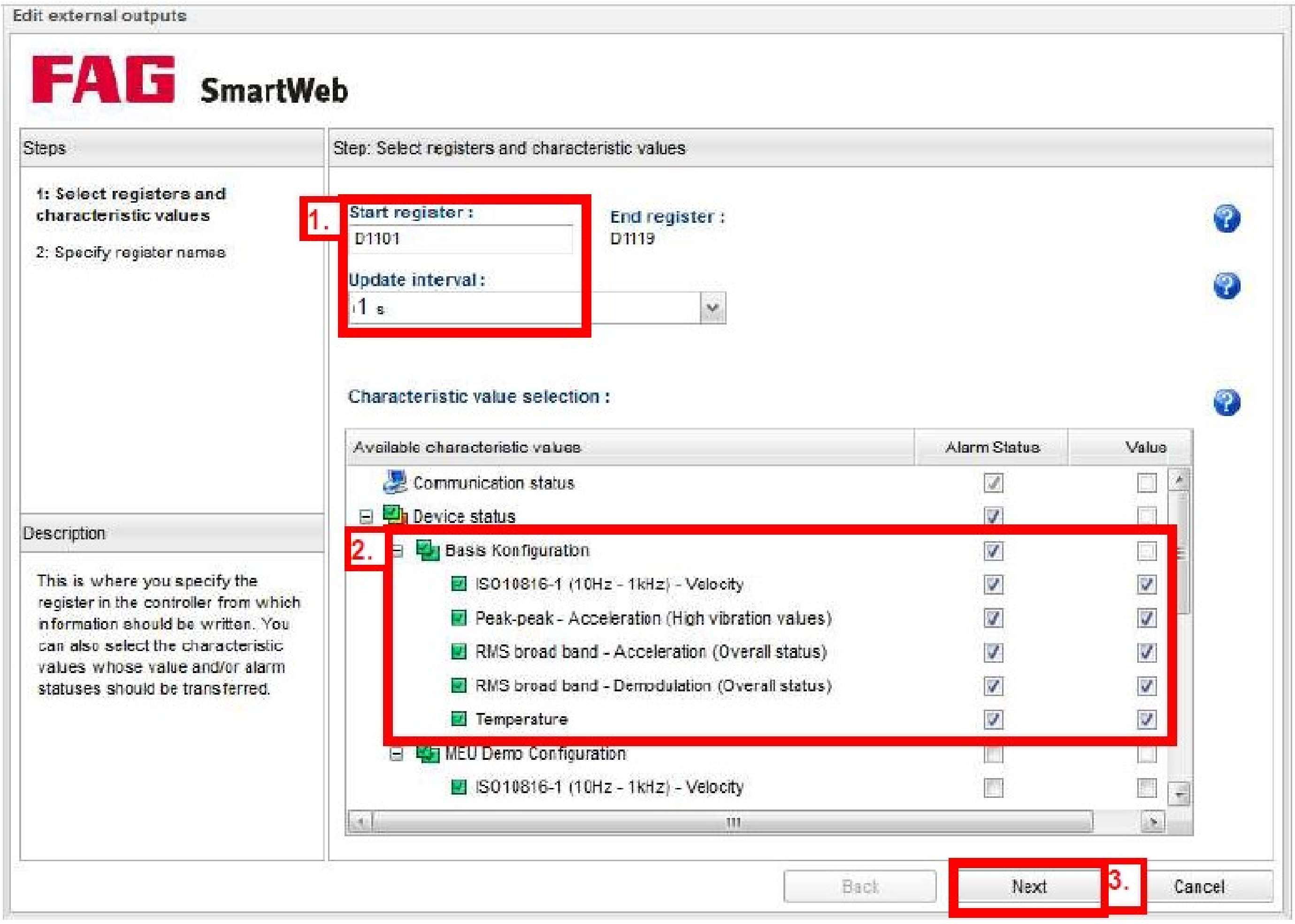


Now we need to specify which D register must be used to the corresponding signals. Select “Configuration“ (1.), “L02“ (2.) and “Create“ at “External outputs for external device: L02“ (3.)

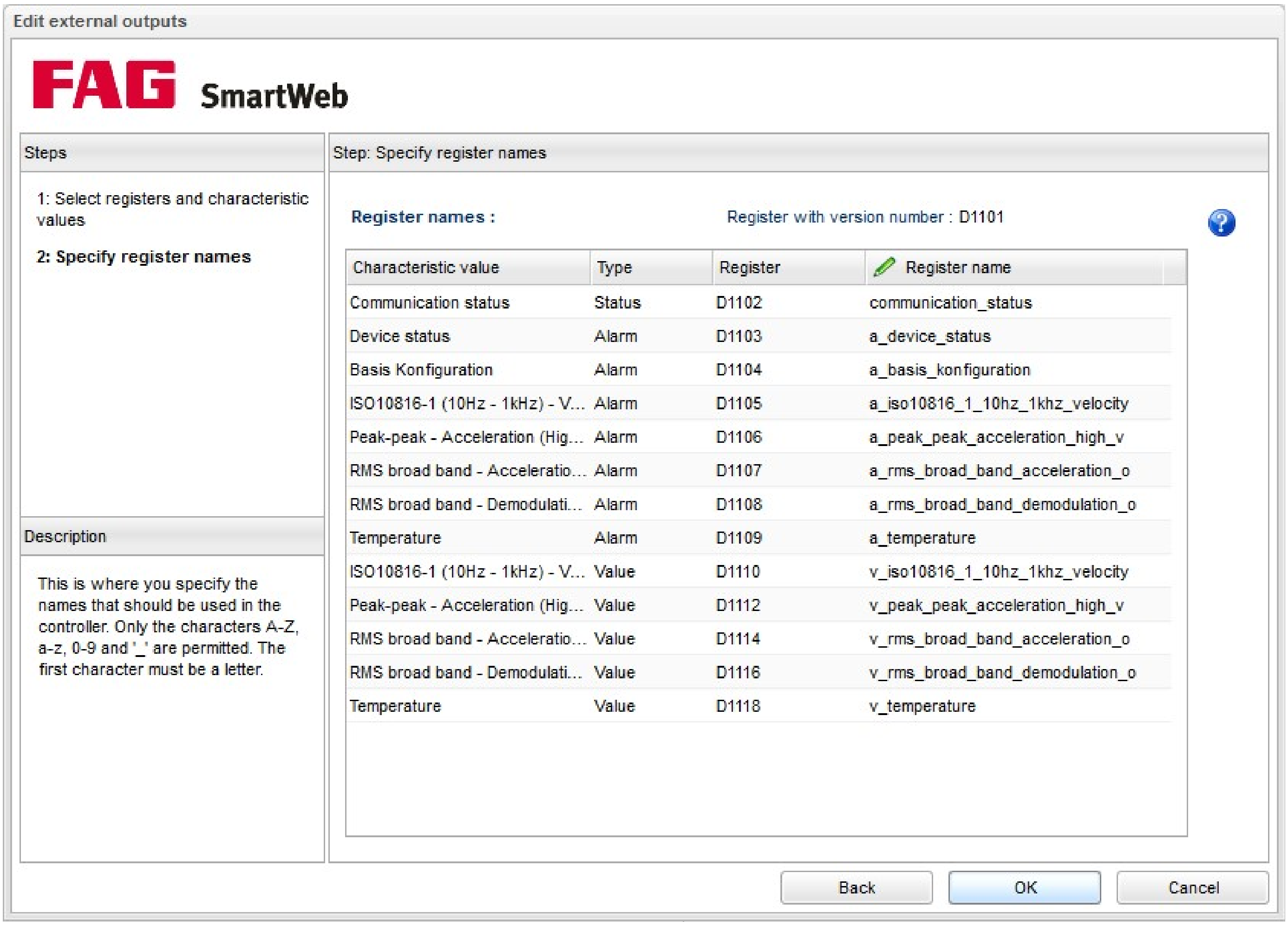


Enter the “Start register” (1.) and “1s” update interval.
Then check all “Alarm Status” and “Value” check boxes as seen below at (2.)

Press “Next” (3.)



In the next upcoming window the register setting can be checked and if everything is correct, then press “OK”



The external device window will then look as follows:

Depending on your MAPS Projects PLC name, MAPS will create a user library inside your GX Works2 project called the same name. lets assume it is called PLC_A, then open the user library called PLC_A inside the MAPS created PLC project and open the POU corresponding to that library. In the POU you will find all your instances of FAG SmartCheck units. Each of the FB instances have an input called “s_cw_Address_offset” and “Config_Version”. Please ensure these inputs corresponds to the “Register with version number” and “Configuration version” as per the image for each individual FAG SmartCheck accordingly. In our case this would have been:

S_cw_Address_Offset = 1101

Config Version = 1.

Faceplate Graphic for the FAG SmartCheck MAPS Template

Analog Input	
	
0000_ABCDE_123	Tag name
Device Status	Shows Status of FAG SmartCheck sensor as follows: GREY – NO COMMS/VALUE ERROR GREEN – NO ALARM/GOOD VALUE YELLOW – PRE-ALARM RED – MAIN-ALARM
Comms Status	Shows Status of FAG SmartCheck Communication to Mitsubishi PLC as follows: GREEN – Good Communications YELLOW – Incorrect Configuration version number in PLC ORANGE – Error reading values from PLC RED – Bad Communications

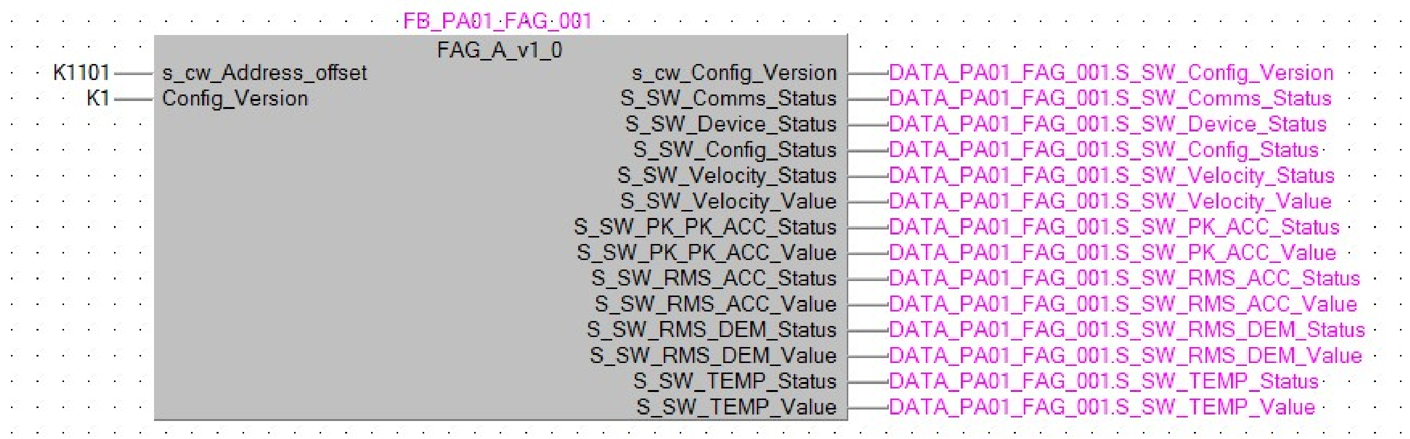
SCADA Agents for the Fag SmartCheck MAPS Template

Signal Description		**Agent Type**
SCADA Status Words:		
SSSL 0	Config Version	**Analog**
SSSL 1	Communications Status	**Marshal**
SSSL 2	Device Status	**Marshal**
SSSL 3	Config Status	**Marshal**
SSSL 4	Velocity Status	**Marshal**
SSSL 5	Peak-to-Peak Acceleration Status	**Marshal**
SSSL 6	RMS Acceleration Status	**Marshal**
SSSL 7	RMS Demodulation Status	**Marshal**
SSSL 8	Temperature Status	**Marshal**
SSSL 9	Velocity Value	**Analog**
SSSL 10	Peak-to-Peak Acceleration Value	**Analog**
SSSL 11	RMS Acceleration Value	**Analog**
SSSL 12	RMS Demodulation Value	**Analog**
SSSL 13	Temperature Value	**Analog**

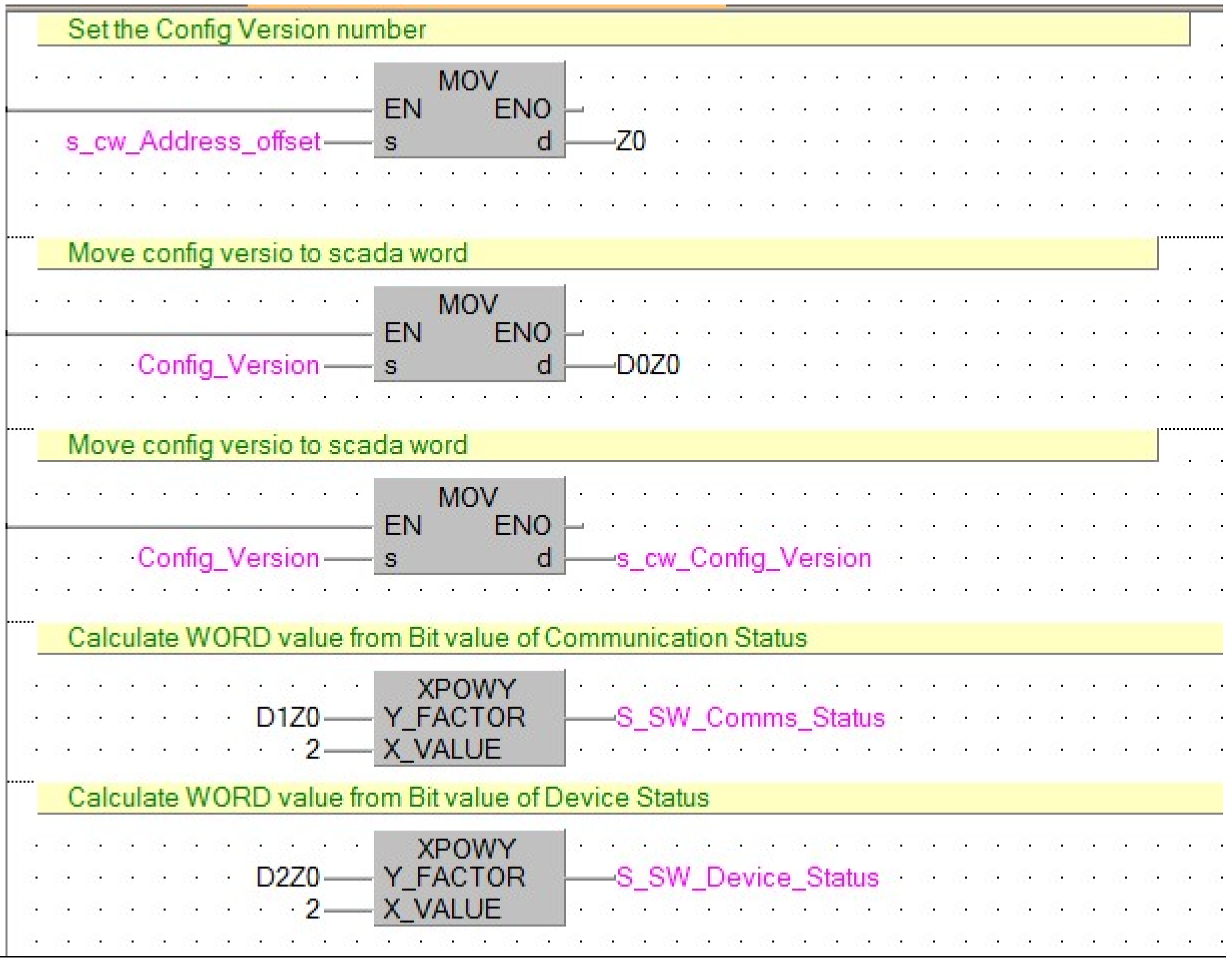
Communications Status Word (Adroit Marshal Agent)		Alarmed
Bit 0	Communications Status OK	
Bit 1	Communication Error. Incorrect Config.	x
Bit 2	Communication Error. No Communications.	x
Bit 3	Communication Error. Alarm States not updated.	x
Bit 4	Spare	
Bit 5	Spare	
Bit 6	Spare	
Bit 7	Spare	
Bit 8	Spare	
Bit 9	Spare	
Bit 10	Spare	
Bit 11	Spare	
Bit 12	Spare	
Bit 13	Spare	
Bit 14	Spare	
Bit 15	Spare	

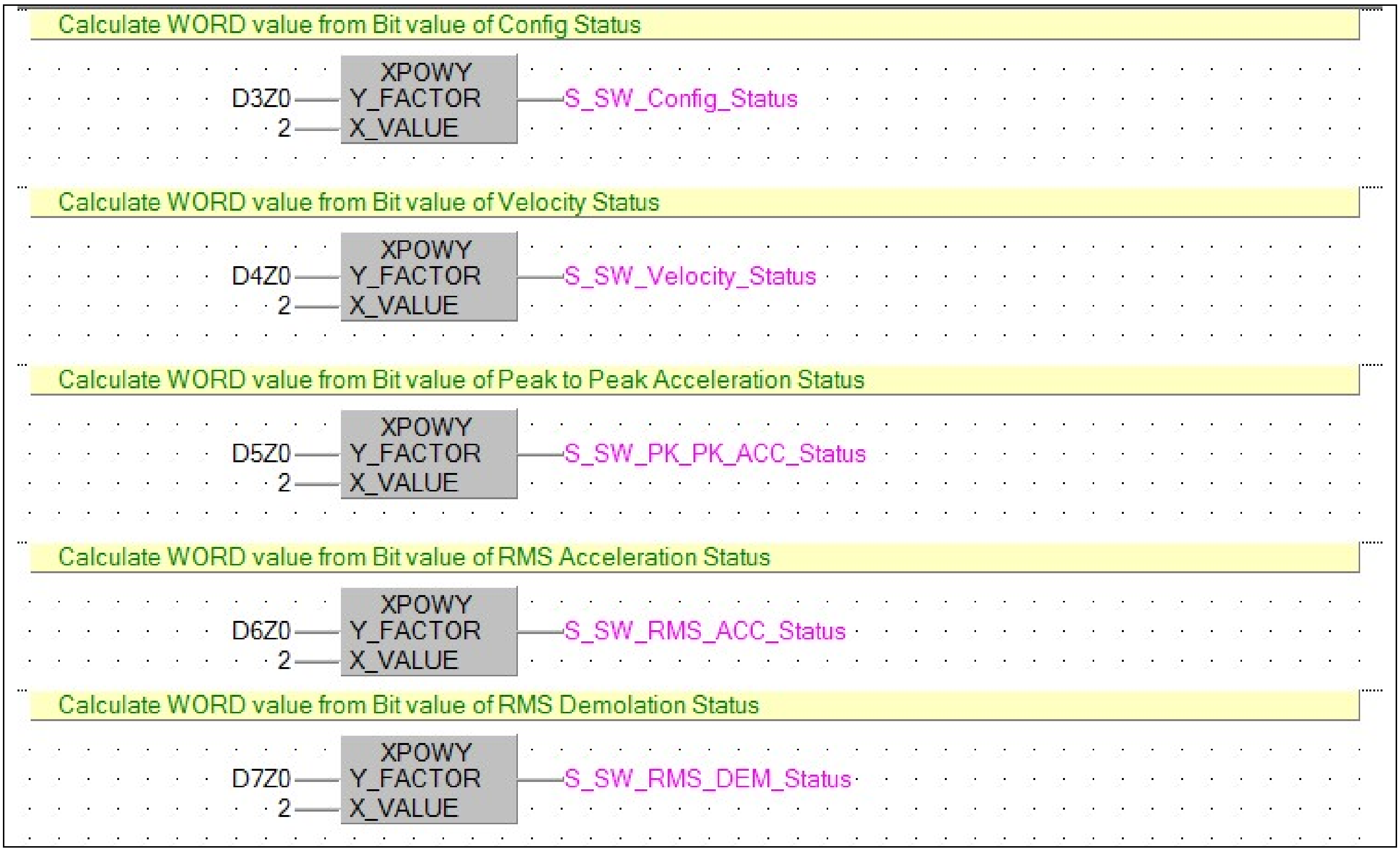
Device\Config\Signal Status Word (Adroit Marshal Agent)		Alarmed
Bit 0	No Comms	x
Bit 1	No Alarm	
Bit 2	Pre-Alarm	x
Bit 3	Alarm	x
Bit 4	No Good Value	X
Bit 5	Good Value	
Bit 6	Spare	
Bit 7	Spare	
Bit 8	Spare	
Bit 9	Spare	
Bit 10	Spare	
Bit 11	Spare	
Bit 12	Spare	
Bit 13	Spare	
Bit 14	Spare	
Bit 15	Spare	
Resources		**Count**
Digital Inputs		0
Digital Outputs		0
Analog Inputs		0
Analog Outputs		0
PLC Memory Steps		359
System Words Used		24
System Bits Used		35
SCADA SCAN Points		14

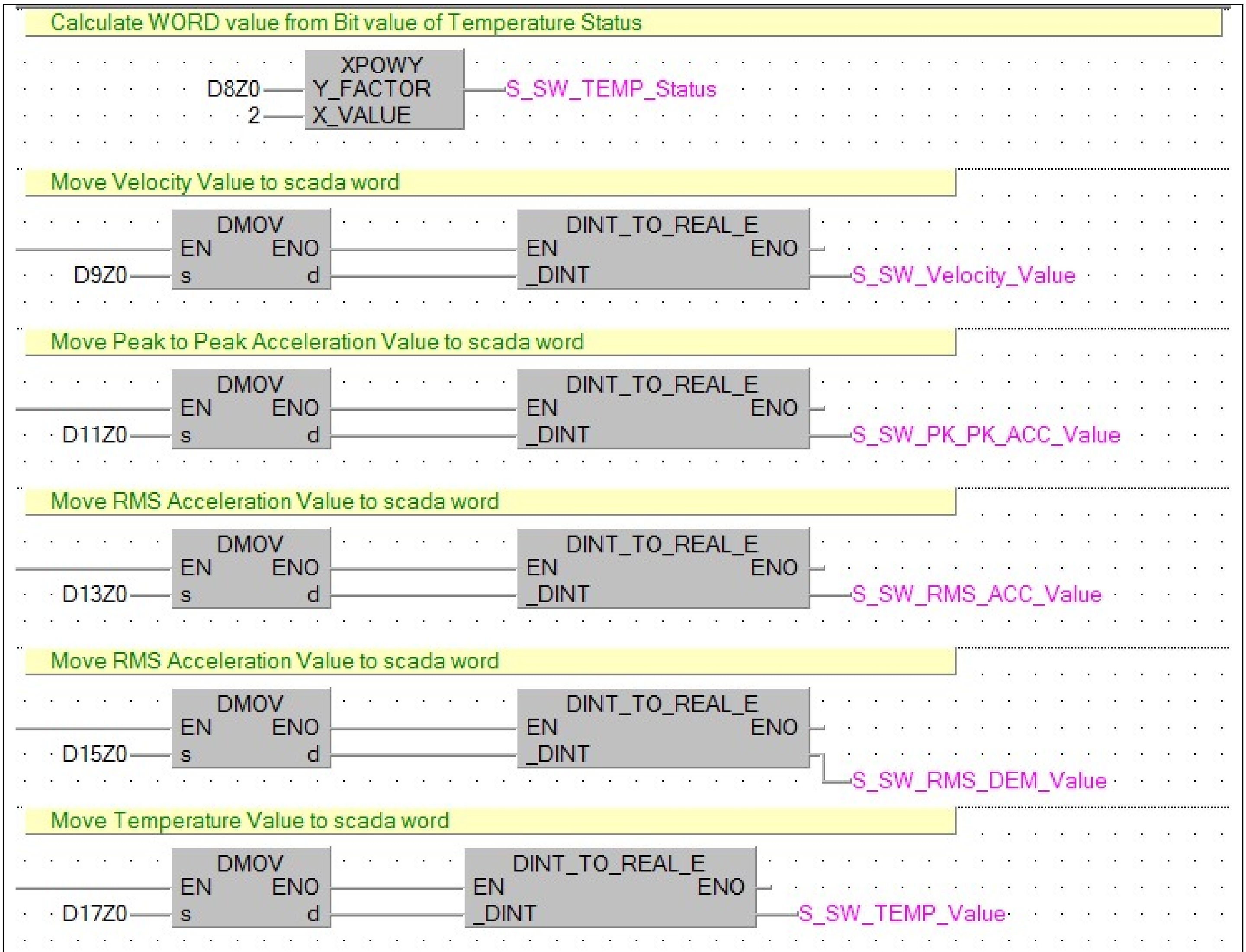
Template: FAG SmartCheck PLC Function Block Instance

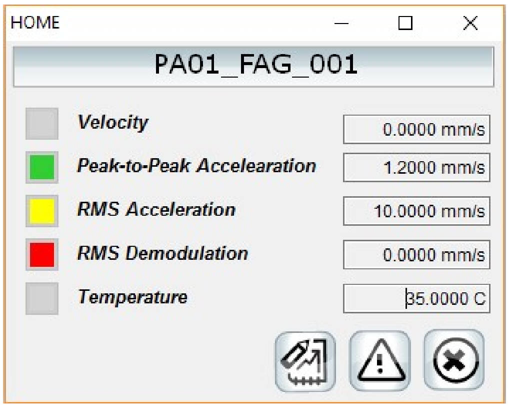
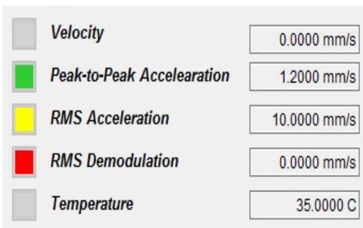


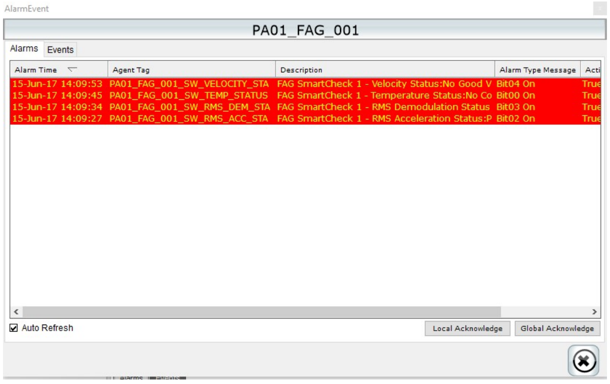
Template: FAG SmartCheck PLC Function Block




Calculate WORD value from Bit value of Config Status
XPOWY
D3Z0
Y_FACTOR
2
X_VALUE
S_SW_Config_Status
Calculate WORD value from Bit value of Velocity Status
XPOWY
D4Z0
Y_FACTOR
2
X_VALUE
S_SW_Velocity_Status
Calculate WORD value from Bit value of Peak to Peak Acceleration Status
XPOWY
D5Z0
Y_FACTOR
2
X_VALUE
S_SW_PK_PK_ACC_Status
Calculate WORD value from Bit value of RMS Acceleration Status
XPOWY
D6Z0
Y_FACTOR
2
X_VALUE
S_SW_RMS_ACC_Status
Calculate WORD value from Bit value of RMS Demolation Status
XPOWY
D7Z0
Y_FACTOR
2
X_VALUE
S_SW_RMS_DEM_Status



Template Graphic: Advanced Home Screen	**Control**	**Descriptions**
	# Equipment ID	Equipment Name
		Signal Status indicators and Values: GREY – NO COMMS/VALUE ERROR GREEN – NO ALARM/GOOD VALUE YELLOW – PRE-ALARM RED – MAIN-ALARM

<table>
<tr><th>Template Graphic: Advanced Alarms & Events Screen</th><th>Control</th><th>Descriptions</th></tr>
<tr><td rowspan="4">
</td><td>Alarms</td><td>Display all the alarms filtered for the current Digital Output</td></tr>
<tr><td>Events</td><td>Display all the events filtered for the current Digital Output.</td></tr>
<tr><td>Local Acknowledge</td><td>Acknowledge all the alarms on locally on the current machine</td></tr>
<tr><td>Global Acknowledge</td><td>Acknowledge all the alarms on globally on all the machines.</td></tr>
</table>

Template Graphic: Advanced Trend Screen